VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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PROJECT SYNOPSIS

Wifi Traffic Anomaly Detection Using Isolation Algorithm

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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2026-2027

1. ABSTRACT

With the rapid growth of wireless communication, WiFi networks have become an essential part of modern digital infrastructure. However, increasing cyber threats such as unauthorized access, network misuse, and abnormal traffic behavior pose serious security challenges. Traditional rule-based monitoring systems often fail to detect unknown or evolving attacks effectively.

This project presents a Real-Time WiFi Traffic Anomaly Detection System using the Isolation Forest algorithm, a machine learning technique for anomaly detection. The system monitors WiFi traffic parameters such as packet size and transmission duration and identifies abnormal patterns in real time. A web-based dashboard built using Streamlit provides live monitoring, visualization, and anomaly highlighting. The system enables early detection of suspicious activities, improves network reliability, and enhances cybersecurity measures in organizations and public networks. The proposed solution is lightweight, scalable, and suitable for real-world deployment.

2. INTRODUCTION

In today's digital era, wireless networks are widely used in homes, organizations, educational institutions, and public places. However, the rapid growth of WiFi usage has also increased the risk of cyber-attacks, unauthorized access, and abnormal network activities. Traditional rule-based monitoring systems are often unable to detect unknown or zero-day attacks effectively.Machine learning techniques provide intelligent solutions for identifying unusual patterns in network traffic. This project aims to design and implement a Real-Time WiFi Traffic Anomaly Detection System using the Isolation Forest algorithm. The system monitors WiFi traffic parameters and identifies abnormal behavior in real time, thereby improving network security, reliability, and performance.

3. MOTIVATION BEHIND THE PROJECT WORK

The increasing number of cyber threats such as Distributed Denial of Service (DDoS), unauthorized access, packet flooding, and network misuse highlights the need for intelligent monitoring systems. Traditional signature-based intrusion detection systems fail to detect new or unknown attack patterns.With the advancement of machine learning techniques, anomaly detection models can identify suspicious behavior without prior knowledge of attack signatures. This project is motivated by the need to develop a lightweight, real-time, and automated WiFi monitoring system that enhances network security in educational institutions, offices, and public hotspots.

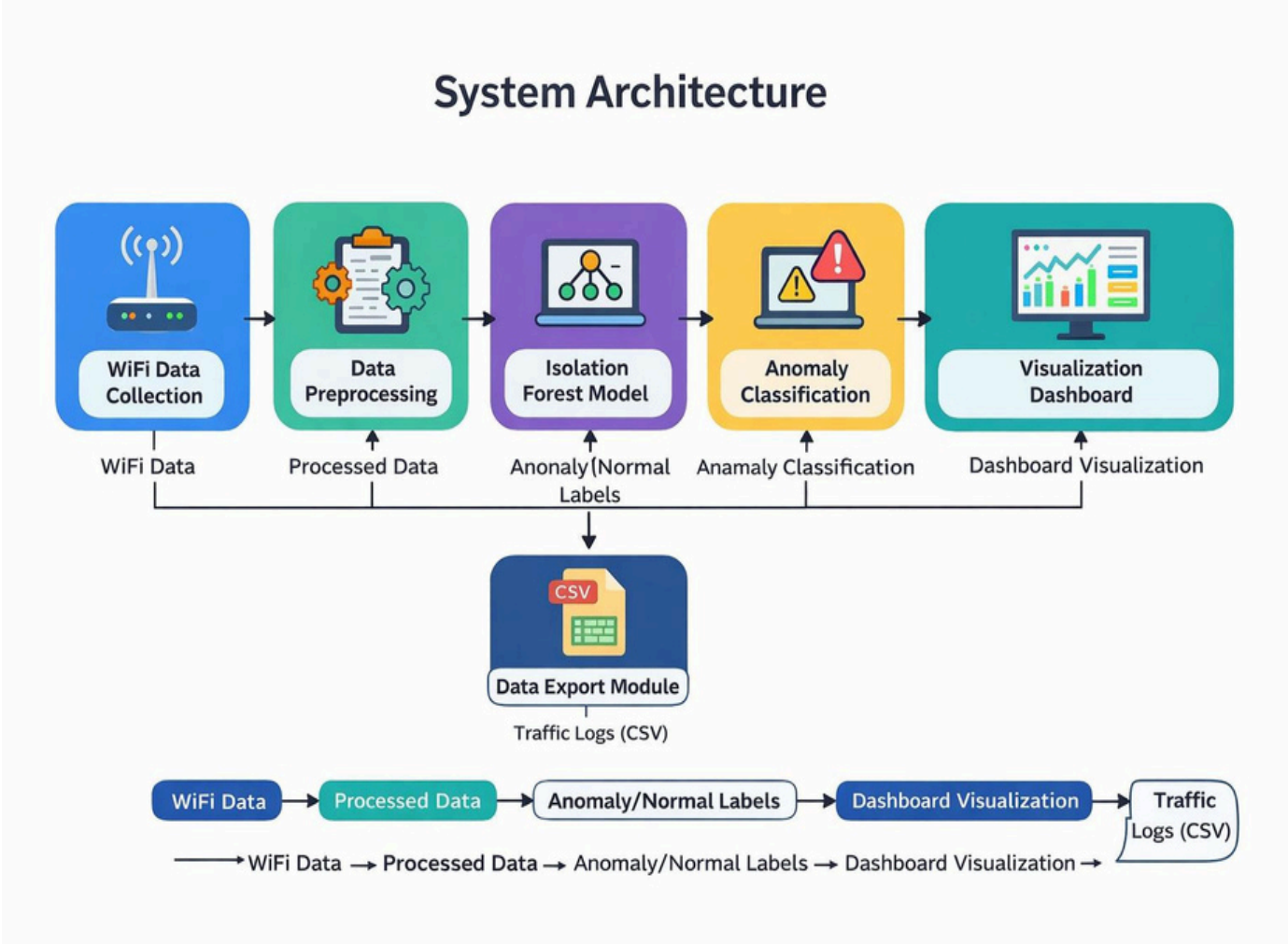
4. PROBLEM STATEMENT DEFINITION

Existing WiFi monitoring systems primarily rely on manual inspection or predefined rules, which are inefficient in detecting unknown or evolving cyber threats. Moreover, small organizations and public networks lack affordable and intelligent monitoring solutions.The problem is to develop a real-time WiFi traffic anomaly detection system that can analyze network traffic parameters and automatically detect abnormal behavior using a machine learning-based approach, specifically the Isolation Forest algorithm.

5. STATEMENT OF OBJECTIVES

- To design a real-time WiFi traffic monitoring system.
- To collect and analyze WiFi traffic parameters such as packet size and transmission duration.
- To implement the Isolation Forest algorithm for anomaly detection.
- To develop an interactive web dashboard using Streamlit for visualization and monitoring.
- To highlight abnormal network behavior for quick identification.
- To export traffic logs for further security analysis.
- To evaluate system performance in detecting abnormal traffic patterns.

6. SYSTEM ARCHITECTURE



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